How much of a ClinVar AUROC is the consequence class?

Charts and checks on the EVEE manuscript's published benchmark data

Tim Richardson
Rewire

2 November 2026

Abstract

Variant effect predictors are usually compared by one area under the ROC curve (AUROC) over ClinVar. EVEE, a probe trained on Evo 2's internal states, reports an AUROC of 0.997 on 833,970 ClinVar single-nucleotide variants (SNVs), and its authors warn that this figure is largely confounded by consequence type. We redrew two of their figures for a talk and checked them against the data they released. We add no new model scores. From the published counts, the share of variants labelled pathogenic runs from 0.10% (95% interval 0.09% to 0.11%) for synonymous to 98.6% (98.4% to 98.8%) for splice donor variants. A score that knows only the consequence class reaches an AUROC of 0.967 (0.967 to 0.968) on the full set. On EVEE's deconfounded set of 158,616 variants it falls to 0.737 (0.735 to 0.739). Within each consequence class, AUROCs are lower and the order of methods changes. On the deconfounded set EVEE leads CADD v1.7 on intronic variants (0.965 against 0.904) and trails REVEL and AlphaMissense on missense variants (0.967 against 0.981 and 0.972). The deconfounded table has no intervals and the methods were scored on different subsets, so we give approximate intervals and make no significance claims there. On the full set, where EVEE published 95% intervals, EVEE's AUROC is above CADD v1.7's in four of the six classes we chart with intervals that do not overlap. Both charts reproduce the published values exactly. The practical point for a laboratory: read a predictor's per-consequence results, not its headline AUROC.

1 Introduction

A variant effect predictor gives each variant a score. Papers usually summarise it with one AUROC over a labelled set, most often ClinVar[1]. AUROC is the chance that a randomly chosen pathogenic variant scores higher than a randomly chosen benign one: 0.5 is chance and 1 is perfect.

ClinVar is not a random sample of variants. Whether a variant is labelled pathogenic depends strongly on its consequence. Nonsense and canonical splice variants are mostly pathogenic; synonymous and intronic variants are almost all benign. A score that has learned consequence, directly or through features such as conservation, can reach a high AUROC without telling apart variants of the same type. Lu and colleagues showed that this kind of heterogeneity inflates reported performance[2]. Grimm and colleagues described related circularity in missense benchmarks[3].

A clinical laboratory gets little from that part of the AUROC. Under ACMG/AMP the consequence already enters the classification on its own terms: a null variant in a loss-of-function gene can meet PVS1, and a computational prediction adds PP3 or BP4 on top[4]. What the lab needs from a predictor is to separate variants within a consequence class.

EVEE is a probe trained on the internal states of the Evo 2 DNA model[5] to predict ClinVar

labels[6]. Its authors report an overall SNV AUROC of 0.997 and say plainly that this figure is "largely confounded by a model's ability to recover consequence-type priors rather than variant-level discrimination"[6]. They built a deconfounded subset and compared methods within each consequence class, and they released the figure data[7].

For a talk on genomic foundation models in rare-disease diagnosis we redrew two of their figures in the talk's colours. This paper documents those charts and the checks we ran on them. It asks three questions:

1. How strongly does consequence predict the ClinVar label in EVEE's sets? We answer with the pathogenic share per class and with the AUROC of consequence alone.
2. What do the within-consequence comparisons on the deconfounded set show, and how uncertain are they?
3. Do our charts show the published values?

This is not a new benchmark. Every model score comes from EVEE's released tables. The only values we compute are derived from their counts and AUROCs, and we mark them as ours.

2 Methods

2.1 Data

All inputs are three files in `artifacts/` of the goodfire-ai/evee-manuscript repository at commit `bb3bfa6` (16 October 2026)[7], read from a local clone and not copied into this repository:

- `dataset_characterization.feather`: for each consequence class, the number of variants and the number labelled pathogenic, in two sets. "Naive (834K)" is the full SNV set; "Deconf (159K)" is the deconfounded set.
- `deconf_benchmark.feather`: for each method and consequence class on the deconfounded set, the number of variants, the number pathogenic, the number the method scored (`n_valid`), AUROC and AUPRC. It has no interval columns.
- `snv_benchmark.feather`: the same for the full SNV set, with a 95% interval for most AUROCs (`auroc_lo`, `auroc_hi`; missing for some sparsely scored rows, such as REVEL on synonymous and nonsense variants). The repository does not include the code that computed these intervals, so we do not know the method.

The EVEE repository describes how the sets were made[7]. Labels come from ClinVar release 2026-03-21 on GRCh38 with GENCODE v49: pathogenic or likely pathogenic against benign or likely benign, at least one review star, genes no longer than 100 kb, with multiallelic, multinucleotide and unannotated entries removed. The deconfounded subset reduces label imbalance within consequence types "while accounting for their prevalence" (README), with variants selected round-robin over genes; the preprint's Methods give the details[6]. It is not a scaled-down copy of the full set: missense variants are 15% of the full set and 64% of the deconfounded set, synonymous variants 35% and 8% (Table 1). The docstring of the repository's Figure 1c script calls this set "CADD-deconfounded v3". We could not find out from the repository whether CADD scores were used in selecting it.

2.2 How EVEE scores a variant

Figure 1 summarises the EVEE probe from the repository's protocol[7]. Evo 2 7B reads a 65 kb window around the variant, for the reference and the variant sequence, on both strands. The probe takes the hidden state of block 27, keeps the 256 positions per strand where reference and variant differ most by cosine divergence, and forms variant minus reference features. A covariance pooling layer and a linear readout, both trained on ClinVar SNV labels, give the

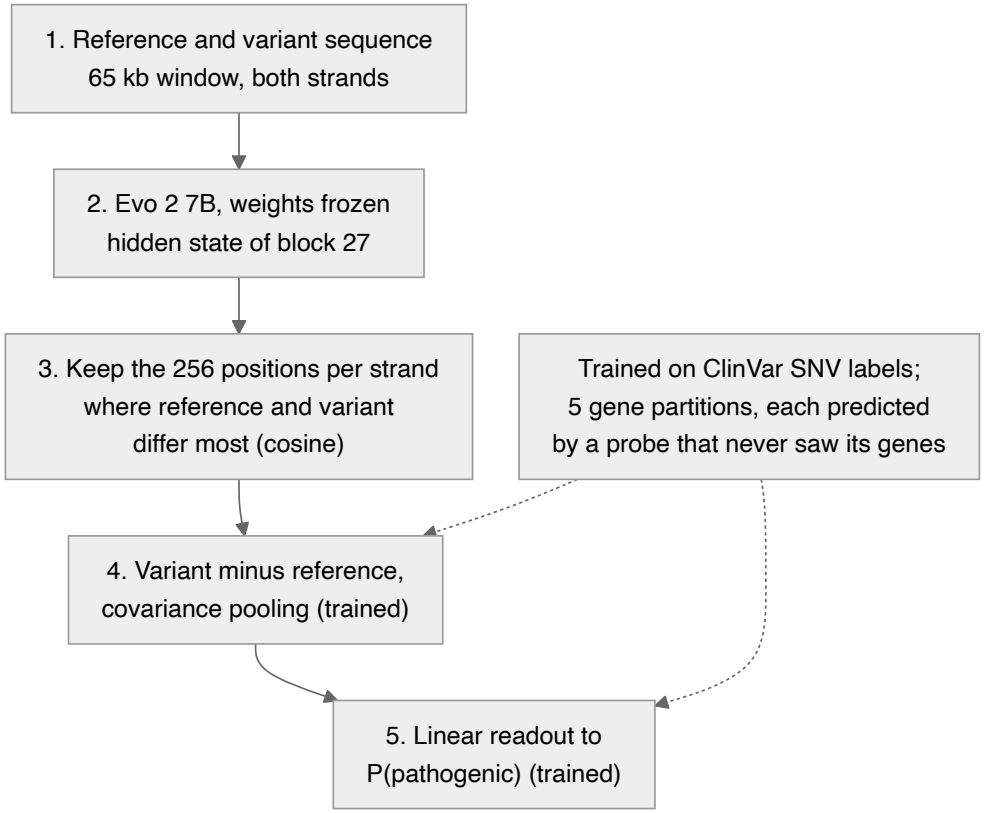


Figure 1. How EVEE scores one SNV. Steps 2 and 3 use Evo 2 unchanged; steps 4 and 5 are trained on ClinVar labels with genes held out. From the evee-manuscript README[7].

probability that the variant is pathogenic. Genes were split into five partitions; each partition was predicted by a probe trained on the other four. The chart label "EVEE probe" is the row named "Covpool probe" in the tables. The tables also have an "Evo2 probe" row, a mean-pooled probe, which we do not chart.

2.3 Methods compared

The results chart shows five of the table's methods: the EVEE probe; Evo 2 zero-shot ("Evo2 loss"), the change in Evo 2's sequence likelihood when the variant is swapped in[5]; AlphaMissense[8]; REVEL[9]; and CADD v1.7[10]. It shows six consequence classes: missense, synonymous, intronic, splice donor, splice acceptor and nonsense. A point is drawn only where the method scored at least 100 variants in that class. A method that cannot score a variant is left out of that class, not counted as wrong, so each AUROC describes the variants that method scored.

2.4 The charts

`evee/evee_charts.py` draws both charts with matplotlib 3.11.2, pandas 3.1.0 and pyarrow 26.0.0 under Python 3.11.13, run by uv 0.8.2:

```
uv run --python 3.11 --with matplotlib==3.11.2 --with pandas==3.1.0 \
    --with pyarrow==26.0.0 \
    python evee/evee_charts.py --evee ../evee-manuscript/artifacts
```

The confounding chart (Figure 3) plots the pathogenic share of eight classes in the full set. The

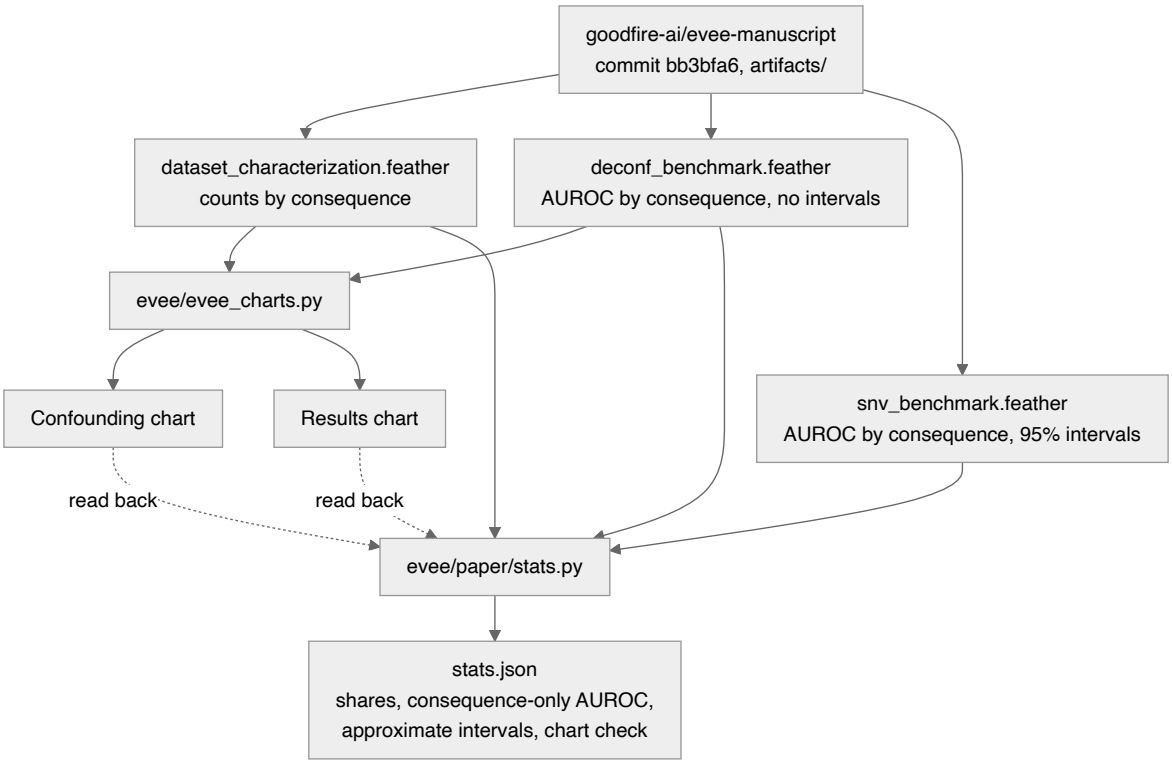


Figure 2. What this folder does with EVEE's released files.

results chart (Figure 4) plots AUROC by class on the deconfounded set. The script writes SVG to `evee/out/`. `paper/make_figures.sh` reruns it, converts the SVGs to PDF and PNG with rsvg-convert 2.62.3, runs `paper/stats.py`, then renders the diagrams with mmdc 11.16.0. Figure 2 shows the flow.

2.5 Values computed here

`evee/paper/stats.py` (pandas 3.1.0, pyarrow 26.0.0 and numpy 2.4.6, same Python) writes every value below to `evee/paper/stats.json`. It needs no GPU and runs in seconds on a laptop.

- **Pathogenic share** per class, with a Wilson 95% interval[11] from the published counts.
- **Consequence alone as a score.** Each variant gets the pathogenic share of its class in that set. The AUROC of this score follows exactly from the counts, with ties counted as one half. The 95% interval comes from 2,000 bootstrap draws (seed 20261102) that resample the pathogenic and the benign variants separately across classes and recompute the shares in each draw. The shares are fitted on the same data they are tested on. To measure the optimism from this, we also compute a two-fold cross-fitted AUROC: each class's pathogenic and benign counts are split at random into halves, the shares are fitted on one half and scored on the other, both ways, and the result is averaged over 200 splits (seed 20261103).
- **Pairs within a class and majority-rule accuracy.** From the same counts: the share of pathogenic and benign pairs that fall in the same consequence class, and the accuracy of predicting each class's majority label.
- **Approximate intervals for the deconfounded AUROCs**, by the Hanley and McNeil formula[12] from the number of pathogenic and benign variants each method scored. Where a method scored only part of a class, the table does not say how many of the scored variants

were pathogenic. We then assume the class's pathogenic share and mark the interval as assumed. This affects REVEL, CADD v1.7 on missense and synonymous (8 and 1 variants unscored) and the overall rows of AlphaMissense and CADD. For AlphaMissense on missense, the scored count equals the number of canonical missense variants, whose split is published, so we use that.

- **Chart check.** The script reads the SVGs back. For the results chart it converts each marker's horizontal position to AUROC using the axis ticks at 0.70 and 1.00, and compares it with the table. For the confounding chart it compares the printed labels with the counts.

2.6 Statistics

No hypothesis was set in advance; every comparison here is exploratory. On the full SNV set we call a difference between two methods significant at the 5% level when their published 95% intervals do not overlap. We apply this rule only between methods that scored the same variants (EVEE, Evo 2 zero-shot and CADD v1.7). For two independent or positively correlated estimates this rule is stricter than a 5% test. Overlapping intervals do not show that there is no difference. We make no correction for multiple comparisons. On the deconfounded set we make no significance claims: the intervals are approximate, and the methods were scored on different, overlapping subsets, so their estimates are neither paired nor independent.

3 Results

3.1 Consequence predicts the label

Table 1 gives the pathogenic share by class in both sets. In the full set, nonsense, splice donor and splice acceptor variants are 98.2% to 98.6% pathogenic. Synonymous variants are 0.10% pathogenic and intronic variants 0.68%. Intronic and synonymous variants make up 547,106 of the 833,970 SNVs. Figure 3 shows eight of the classes.

Table 1. Pathogenic share by consequence in EVEE's two ClinVar SNV sets. Counts from `dataset_characterization.feather`; Wilson intervals computed here. All: 96,793 and 44,057 pathogenic.

Consequence	Full set: n	Pathogenic, % (95% CI)	Deconfounded: n	Pathogenic, % (95% CI)
Splice donor	11,855	98.6 (98.4 to 98.8)	1,853	92.4 (91.2 to 93.6)
Nonsense	36,050	98.3 (98.2 to 98.5)	6,367	91.4 (90.7 to 92.1)
Splice acceptor	9,828	98.2 (98.0 to 98.5)	1,650	90.7 (89.2 to 92.0)
Initiator codon	1,363	87.4 (85.5 to 89.0)	715	78.2 (75.0 to 81.1)
Missense	125,092	29.0 (28.8 to 29.3)	101,666	32.0 (31.7 to 32.3)
Other	3,498	7.4 (6.6 to 8.3)	187	54.5 (47.4 to 61.5)
Intronic	254,737	0.68 (0.65 to 0.71)	27,644	4.95 (4.70 to 5.21)
5' UTR	15,232	0.54 (0.43 to 0.67)	1,480	5.07 (4.06 to 6.31)
3' UTR	16,679	0.16 (0.11 to 0.24)	876	2.85 (1.94 to 4.18)
Non-coding transcript	67,267	0.16 (0.13 to 0.19)	3,593	2.70 (2.22 to 3.28)
Synonymous	292,369	0.10 (0.09 to 0.11)	12,585	2.30 (2.05 to 2.57)
All	833,970	11.6	158,616	27.8

A score that knows only the consequence class reaches an AUROC of 0.967 (0.967 to 0.968) on the full set. That is above Evo 2 zero-shot on the same set (0.940, 0.939 to 0.941). On the

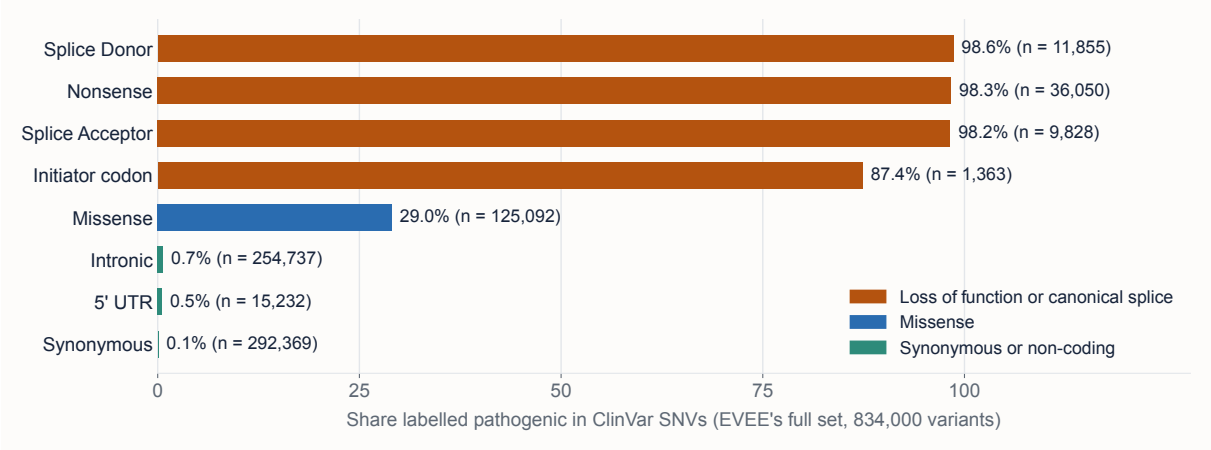


Figure 3. Share of variants labelled pathogenic, by consequence, in EVEE's full set of 833,970 ClinVar SNVs. Drawn from `dataset_characterization.feather`.

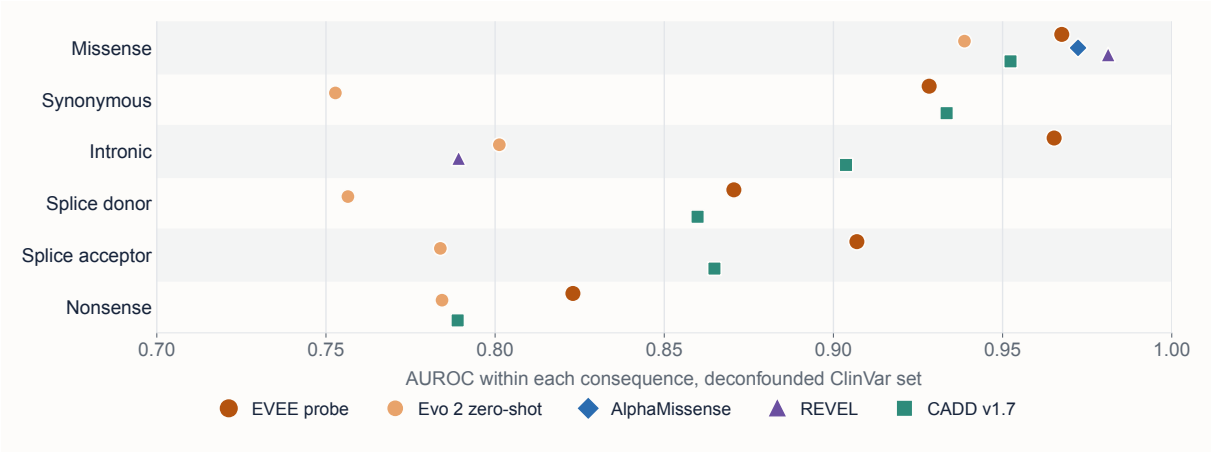


Figure 4. AUROC within each consequence class on EVEE's deconfounded ClinVar set. Points where a method scored fewer than 100 variants are left out. Drawn from `deconf_benchmark.feather`.

deconfounded set, consequence alone gives 0.737 (0.735 to 0.739). Fitting the shares on half the data and scoring the other half gives 0.967 and 0.737 again (0.96718 and 0.73675 against 0.96720 and 0.73680 in-sample), so the in-sample fit adds less than 0.0001. The preprint reports a similar fall in consequence-only accuracy, from 96% to 73% (Methods, "Clinical variant datasets")[6]; its exact rule is not given, and a majority rule per class on the released counts gives 95.2% and 77.7%. The deconfounding narrows the gaps between classes but does not remove them: pathogenic shares still run from 2.30% (synonymous) to 92.4% (splice donor).

3.2 Within each consequence, the deconfounded set

Figure 4 is the results chart and Table 2 gives its values. No method reaches the overall AUROC of 0.997 in any of these classes. CADD v1.7 ranges from 0.789 (nonsense) to 0.952 (missense) within a class, against 0.963 overall on this set.

Table 2. AUROC on the deconfounded set, from `deconf_benchmark.feather`, with approximate 95% intervals computed here (Hanley and McNeil). * The pathogenic count among scored variants is not published; the interval assumes the class's share. Methods with fewer than 100 variants scored in a class are left out, as on the chart.

Class	Method	AUROC	Approximate 95% CI	Variants scored
Missense	EVEE probe	0.967	0.964 to 0.971	20,679
	Evo 2 zero-shot	0.939	0.937 to 0.941	101,666
	AlphaMissense	0.972	0.971 to 0.974	95,791
	REVEL	0.981	0.979 to 0.983*	29,113
	CADD v1.7	0.952	0.951 to 0.954*	101,658
Synonymous	EVEE probe	0.928	0.879 to 0.978	2,557
	Evo 2 zero-shot	0.753	0.720 to 0.786	12,585
	CADD v1.7	0.933	0.913 to 0.953*	12,584
Intronic	EVEE probe	0.965	0.951 to 0.980	5,771
	Evo 2 zero-shot	0.801	0.787 to 0.815	27,644
	REVEL	0.789	0.548 to 1.000*	106
	CADD v1.7	0.904	0.893 to 0.915	27,644
Splice donor	EVEE probe	0.871	0.825 to 0.916	391
	Evo 2 zero-shot	0.756	0.722 to 0.791	1,853
	CADD v1.7	0.860	0.837 to 0.883	1,853
Splice acceptor	EVEE probe	0.907	0.871 to 0.943	377
	Evo 2 zero-shot	0.784	0.753 to 0.815	1,650
	CADD v1.7	0.865	0.843 to 0.887	1,650
Nonsense	EVEE probe	0.823	0.793 to 0.853	1,320
	Evo 2 zero-shot	0.784	0.768 to 0.800	6,367
	CADD v1.7	0.789	0.773 to 0.805	6,367

What the table shows, with no significance claims (see Statistics):

- **Intronic.** EVEE 0.965 against CADD v1.7 0.904. The approximate intervals are well apart. This is the largest gap to CADD in the chart.
- **Missense.** EVEE 0.967 is below REVEL (0.981) and AlphaMissense (0.972) and above CADD v1.7 (0.952). The approximate intervals are apart for REVEL and CADD v1.7. For AlphaMissense they almost touch (EVEE up to 0.9706, AlphaMissense from 0.9710). EVEE's missense AUROC is on 20,679 variants, AlphaMissense's on 95,791 and REVEL's on 29,113.
- **Splice donor, splice acceptor, nonsense, synonymous.** EVEE is above CADD v1.7 in the first three and just below it on synonymous variants (0.928 against 0.933). Every one of these pairs of intervals overlaps. EVEE's subsets here are small: 391, 377, 1,320 and 2,557 variants, with only 50 pathogenic synonymous variants.
- **EVEE against Evo 2 zero-shot.** The probe is higher in all six classes. The intervals are apart in five; for nonsense they overlap (0.793 to 0.853 against 0.768 to 0.800).
- **REVEL on intronic variants** rests on 106 scored variants, about 5 of them pathogenic if the class share holds. Its interval runs from 0.548 to 1.000. The point on the chart carries almost no information.

The methods were not scored on the same variants. EVEE was scored on 32,437 of the 158,616 deconfounded variants, 20.5% of them; CADD v1.7 on 158,607 and Evo 2 zero-shot on all 158,616. On the full set, by contrast, EVEE has a score for all 833,970 variants. The table's Overall row gives 833,970 scored variants, but its "Other" row for the EVEE probe has 390 variants against 3,498 for every other method, so its class rows sum to 830,862. The six classes we report are

complete. The repository does not explain the difference. A fraction of about one in five would fit a single held-out gene partition, but we could not confirm this.

3.3 Within each consequence, the full set

The full-set table has published 95% intervals and EVEE scores every variant in the six classes we report, so it allows comparisons on nearly the same variants for EVEE, Evo 2 zero-shot and CADD v1.7; AlphaMissense and REVEL scored subsets. Table 3 gives the same five methods and six classes.

Table 3. AUROC on EVEE's full SNV set with the published 95% intervals, from `snv_benchmark.feather`. EVEE and Evo 2 zero-shot scored all variants in each class; CADD v1.7 all but 1,280 of 833,970, nearly all of them in the "Other" class. Rows with fewer than 100 scored variants (AlphaMissense on splice acceptor, 29) and rows without a published interval (REVEL on synonymous and nonsense variants) are left out.

Class	Method	AUROC	Published 95% CI	Variants scored
All SNVs	EVEE probe	0.9968	0.9967 to 0.9970	833,970
	Evo 2 zero-shot	0.9401	0.9393 to 0.9410	833,970
	CADD v1.7	0.9914	0.9911 to 0.9917	832,690
Missense	EVEE probe	0.971	0.970 to 0.972	125,092
	Evo 2 zero-shot	0.926	0.924 to 0.927	125,092
	AlphaMissense	0.968	0.967 to 0.969	117,653
	REVEL	0.978	0.977 to 0.980	36,783
	CADD v1.7	0.946	0.945 to 0.947	125,082
Synonymous	EVEE probe	0.961	0.948 to 0.973	292,369
	Evo 2 zero-shot	0.670	0.643 to 0.700	292,369
	CADD v1.7	0.929	0.909 to 0.950	292,368
Intronic	EVEE probe	0.984	0.980 to 0.987	254,737
	Evo 2 zero-shot	0.801	0.787 to 0.814	254,737
	REVEL	0.771	0.604 to 0.908	752
	CADD v1.7	0.910	0.900 to 0.918	254,736
Splice donor	EVEE probe	0.920	0.894 to 0.944	11,855
	Evo 2 zero-shot	0.762	0.714 to 0.811	11,855
	CADD v1.7	0.857	0.819 to 0.893	11,855
Splice acceptor	EVEE probe	0.928	0.902 to 0.951	9,828
	Evo 2 zero-shot	0.800	0.759 to 0.840	9,828
	CADD v1.7	0.882	0.841 to 0.916	9,828
Nonsense	EVEE probe	0.900	0.885 to 0.914	36,050
	Evo 2 zero-shot	0.809	0.788 to 0.828	36,050
	CADD v1.7	0.783	0.760 to 0.801	36,050

By the rule in Statistics, all exploratory:

- **EVEE against CADD v1.7.** EVEE is higher overall (0.997 against 0.991) and within missense, intronic, splice donor and nonsense variants, with intervals that do not overlap. For splice donor the gap between the intervals is 0.001. On synonymous and splice acceptor variants the intervals overlap: not shown.
- **EVEE against Evo 2 zero-shot.** EVEE is higher overall and in all six classes, with intervals that do not overlap.
- **Missense against the protein tools.** EVEE has 0.971, AlphaMissense 0.968 and REVEL 0.978. AlphaMissense and REVEL scored 117,653 and 36,783 of the 125,092 missense variants, so these AUROCs are on different variants and we do not apply the

overlap rule to them. REVEL is higher on the missense variants it scores.

The overall AUROC compresses large differences. EVEE and CADD v1.7 differ by 0.005 overall, but by 0.025 on missense, 0.074 on intronic and 0.117 on nonsense variants. CADD v1.7 ranges from 0.783 to 0.946 across these six classes against 0.991 overall.

EVEE's within-class AUROCs are higher on the full set than on the deconfounded set: intronic 0.984 against 0.965, splice donor 0.920 against 0.871, nonsense 0.900 against 0.823. CADD v1.7 changes less between the two (intronic 0.910 against 0.904, nonsense 0.783 against 0.789). The two sets differ in composition and EVEE's deconfounded subset is smaller, so we do not interpret this further.

3.4 The charts match the published values

All 21 markers on the results chart decode to the table's AUROC; the largest difference is 2×10^{-10} (`stats.json`, `max_results_diff`). All 8 labels on the confounding chart match the counts to the one decimal place shown.

4 Discussion

Consequence alone reaches an AUROC of 0.967 on EVEE's full ClinVar SNV set. In the full set 95% of pathogenic-benign pairs are in different consequence classes (within-class pairs are 5.3%), so the overall AUROC mostly measures how a score ranks classes. So a headline AUROC of 0.99 or more says little about how well a tool separates two missense variants, or two intronic variants. The overall gap between EVEE and CADD v1.7 is 0.005; within a class it is 0.025 to 0.117 on the full set. Within-class comparisons, which the EVEE authors report themselves, are the numbers a laboratory should read.

The within-class results favour EVEE most outside missense variants. On intronic variants EVEE's lead over CADD v1.7 is large on both sets. On missense variants REVEL is higher on both sets, on the variants it scores (36,783 of 125,092 on the full set), and AlphaMissense is within 0.005 of it on both. The classes outside missense are where a DNA model could add something that protein tools cannot: AlphaMissense has no intronic, synonymous or splice rows at all on the deconfounded set, and REVEL scores only 106 of 27,644 intronic variants.

The probe also beats Evo 2's own zero-shot score in every class. The EVEE authors take this as evidence that Evo 2's internal states carry more than its likelihood does[6]. Part of the gain may come from the labels: the probe is trained on ClinVar and tested on ClinVar, in held-out genes, while CADD v1.7 and the zero-shot score never saw ClinVar labels. The authors note that ClinVar benchmarks favour models trained on ClinVar[6].

4.1 Limits

- **No new scores.** We did not run any model. Every AUROC is EVEE's. Our own work is the charts, the check that they match, and values derived from published counts.
- **Inputs not in this repository.** The three feather files are read from a clone of goodfire-ai/evee-manuscript at commit `bb3bfa6`. If that repository changes, the charts can change with it.
- **No published intervals on the deconfounded set.** Our Hanley and McNeil intervals are approximate. The formula assumes a particular score distribution, and for REVEL and part of CADD the pathogenic count among scored variants is assumed. We make no significance claims on this set.
- **Different subsets.** On the deconfounded set EVEE was scored on 20.5% of the variants and the other methods on more. A tool that cannot score a variant is left out, not counted

as wrong. Two AUROCs in the same row of Table 2 are therefore on different variants. We do not know why EVEE has a full-set score for every variant but a deconfounded score for only a fifth.

- **Unknown interval method on the full set.** The published intervals come without the code that made them.
- **The deconfounded set's construction.** It is described in the preprint, not in code we could run, and its script names it "CADD-deconfounded". If CADD scores were used in selecting it, CADD's AUROCs on it could be biased.
- **Consequence-only AUROC is in-sample.** The shares are fitted on the set they score. The two-fold cross-fitted value is lower by less than 0.0001 on both sets.
- **Exploratory comparisons.** None were planned in advance, and there is no correction for multiple comparisons.
- **ClinVar labels.** The labels are themselves partly based on computational evidence and on consequence, which can favour tools that use the same evidence[3].

5 Conclusion

In EVEE's ClinVar SNV set the consequence class alone gives an AUROC of 0.967. Headline AUROCs near 0.99 mostly measure that. Within each class, AUROCs are lower and the order of methods changes: EVEE leads clearly on intronic variants, REVEL is higher on the missense variants it can score (36,783 of 125,092 on the full set), and on the deconfounded set EVEE's gaps to CADD v1.7 in the synonymous, splice and nonsense classes are within the approximate intervals. Our two charts show EVEE's published values exactly. A lab choosing a predictor should look at its results within the consequence classes that matter for its cases, on variants the tool can actually score.

6 Data and code availability

- **Code.** `evee/evee_charts.py` (the charts), `evee/paper/stats.py` (the values computed here, written to `evee/paper/stats.json`) and `evee/paper/make_figures.sh` are in the repository rewire-bio/amp-2026-studies[13], under the MIT licence. The chart script has been in every release since tag `amp-2026-v4`. This paper, `stats.py`, `make_figures.sh` and the shared build tooling in `tools/paper/` are not yet in a tagged release; the latest tag is `amp-2026-v5`.
- **Rebuild.** With goodfire-ai/evee-manuscript cloned next to the repository at commit `bb3bfa6`, run `bash evee/paper/make_figures.sh`, then `bash tools/paper/build.sh evee/paper`.
- **Inputs.** `dataset_characterization.feather`, `deconf_benchmark.feather` and `snv_benchmark.feather` from goodfire-ai/evee-manuscript at commit `bb3bfa6`[7]. The repository has no licence file at that commit, so we do not copy the files; the counts and AUROCs we use are in `evee/paper/stats.json`. The per-variant EVEE scores on Zenodo (10.5281/zenodo.19701997) are under CC BY 4.0, but we did not use them here. ClinVar, the source of the labels, is in the public domain[1].

References

1. Landrum, M. J. *et al.* ClinVar: Improving access to variant interpretations and supporting evidence. *Nucleic Acids Research* **46**, D1062-D1067 (2018).
2. Lu, B., Liu, X., Lin, P.-Y. & Brandes, N. Genomic heterogeneity inflates the performance of variant pathogenicity predictions. *Nature Communications* https://doi.org/10.1038/s41467-026-78623-5 (2026) doi:10.1038/s41467-026-78623-5.

3. Grimm, D. G. *et al.* The evaluation of tools used to predict the impact of missense variants is hindered by two types of circularity. *Human Mutation* **36**, 513-523 (2015).
4. Richards, S. *et al.* Standards and guidelines for the interpretation of sequence variants: A joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine* **17**, 405-424 (2015).
5. Brixi, G., Durrant, M. G., Ku, J., *et al.* Genome modelling and design across all domains of life with Evo 2. *Nature* **652**, 1349-1361 (2026).
6. Pearce, M. T., Dooms, T., Yamamoto, R., *et al.* Interpretable variant effect prediction from genomic foundation model representations. *bioRxiv* https://doi.org/10.64898/2026.04.10.717844 (2026) doi:10.64898/2026.04.10.717844.
7. Goodfire. evee-manuscript: Code, probe weights and figure data for the EVEE preprint (commit bb3bfa6). (2026).
8. Cheng, J. *et al.* Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science* **381**, eadg7492 (2023).
9. Ioannidis, N. M. *et al.* REVEL: An ensemble method for predicting the pathogenicity of rare missense variants. *American Journal of Human Genetics* **99**, 877-885 (2016).
10. Schubach, M., Maass, T., Nazaretyan, L., Röner, S. & Kircher, M. CADD v1.7: Using protein language models, regulatory CNNs and other nucleotide-level scores to improve genome-wide variant predictions. *Nucleic Acids Research* **52**, D1143-D1154 (2024).
11. Wilson, E. B. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association* **22**, 209-212 (1927).
12. Hanley, J. A. & McNeil, B. J. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology* **143**, 29-36 (1982).
13. Richardson, T. AMP 2026 case tests: Rare disease SNV, indel and inherited cancer panel cases, the 500-case replication, the AlphaGenome AVI lookups, the Evo 2 runs and the reference pipeline. (2026).